

CNSCC.160 – Fundamentals of Communication Systems

Tutorial Questions and Revision (Cryptography: The Science of Secrecy)

Knowledge

Q1. Which of the following attacks is a threat to confidentiality?

- a. snooping
- b. masquerading
- c. repudiation

Only snooping is a threat to confidentiality. Masquerading and repudiation are threats to integrity.

Q2. Which of the following attacks is a threat to integrity?

- a. modification
- b. replaying
- c. denial of service

Replaying and modification are threats to integrity. Denial of service is a threat to availability.

Q3. Which of the following attacks is a threat to availability?

- a. repudiation
- b. denial of service
- c. modification

Only denial of service is a threat to availability. Repudiation and modification are threats to integrity.

Q4. Which of the following words mean 'secret writing'? Which one means 'covered writing'?

- a. cryptography
- b. steganography

Cryptography means "secret writing." Steganography means "covered writing."

Q5. Alice has found a way to write secretly to Bob. Each time, she takes a new text, such as an article from the newspaper, but inserts one or two spaces between the words. A single space means a binary digit 0; a double space means a binary digit 1. Bob extracts the binary digits and interprets them using ASCII code. Is this an example of cryptography or steganography? Explain.

This is an example of steganography because the message is hidden inside the essay.

Q6. Alice and Bob exchange confidential messages. They share a very large number as the encryption and decryption key in both directions. Is this an example of symmetric-key or asymmetric-key cryptography? Explain.

This is an example of symmetric-key cryptography because they share the same secret key and use the same key for communication in both directions, the meaning of symmetric.

Q7. Distinguish between a substitution cipher and a transposition cipher.

In a substitution cipher a set of characters (or bits) in the plaintext is substituted by another set of characters (or bits) in the ciphertext; a transposition cipher changes the positions of the characters (or bits) in the plaintext to create a ciphertext.

Q8. In a cipher, all the As in the plaintext have been changed to Ds in the ciphertext and all the Ds in the plaintext have been changed to Hs in the ciphertext. Is this a monoalphabetic or polyalphabetic substitution cipher? Explain.

This is a monoalphabetic substitution cipher because changing depends on what the character is, not on the position of the character.

Q9. Which cipher can be broken more easily, monoalphabetic or polyalphabetic?

Definitely, a monoalphabetic cipher can be broken more easily. For example, if we find that the decryption of x is a, then all x' s are decrypted as a' s. This is not true with polyalphabetic ciphers.

Q10. Assume Alice and Bob use an additive cipher in modulo 26 arithmetic. If Eve, the intruder, wants to break the code by trying all possible keys (brute-force attack), how many keys should she try on average?

The key in this case needs to be between 0 and 25 (although there is no cipher if the key is 0). on average Eve needs to test $26/2 = 13$ keys to break the code.

Q11. Assume we have a plaintext of 1000 characters. How many keys do we need to encrypt or decrypt the message in each of the following ciphers?

- a. additive
- b. monoalphabetic
- c. autokey

For the additive cipher, we use only one key. For the monoalphabetic cipher, we use 26 keys.

For the autokey cipher, we use 1000 keys.

Q12. According to the definitions of stream and block ciphers, find which of the following ciphers is a stream cipher.

- a. additive
- b. monoalphabetic
- c. autokey

All are examples of stream ciphers. In the additive cipher, the key stream is repeated values of the key. In the monoalphabetic cipher, each key in the stream is a mapping from the plaintext to ciphertext. In the autokey cipher, each key is the previous plaintext character.

Q13. Why do you think asymmetric-key cryptography is used only with small messages?

To be safe, the keys used in asymmetric-key cryptography should be very large. This makes the calculation very long if the message is also long. Compare the short key of DES (56 bits) with the long key of RSA (500 to 1000 bits).

Q14. In an asymmetric public key cipher, which key is used for encryption? Which key is used for decryption?

- a. public key
- b. private key

In an asymmetric public key, the sender uses the public key of the receiver for encryption. The receiver uses her private key for decryption.

Q15. What is the role of the secret key added to the hash function in Figure (on slide 11, week 8, part2) in the text (MAC)? Explain.

The secret key eliminates the need for a secure channel to send the digest. In other words, the secret key prevents Eve from intercepting the digest and changing it to her benefit.

Q16. Distinguish between message authentication and entity authentication.

Message authentication and entity authentication both authenticate the sender for the receiver. As their names indicate, however, message authentication authenticates the sender for a particular message, while entity authentication authenticates the sender for the entire session in which several messages can be sent. If the sender is sending only a single message, message authentication has the same effect as entity authentication; if a sender is sending several messages, one entity authentication authenticates the sender for all messages.

Q17. Alice signs the message she sends to Bob to prove that she is the sender of the message. Which of the following keys does Alice need to use?

- a. Alice's public key
- b. Alice's private key

Alice needs to use her private key to sign the message. Alice cannot sign with her public key, which anyone could use to sign for Alice.

Q18. Alice needs to send a message to a group of 50 people. If Alice needs to use message authentication, which of the following schemes do you recommend?

- a. MAC
- b. digital signature

The first choice is actually very inefficient and not practical because Alice needs to have a shared secret key between herself and each of the recipients. She also needs to create fifty separate MACs, one for each secret key. Alice can use digital signature and sign only one message with her own private key. She then sends a copy of the message to each recipient. Each recipient needs to use Alice's public key to verify the message.

Q19. Which of the following services are not provided by digital signature?

- a.** message authentication
- b.** confidentiality
- c.** nonrepudiation

Digital signature cannot provide confidentiality.

Understanding

Q1. Define the type of attack in each of the following cases:

- a. A student breaks into a professor's office to obtain a copy of the next test.
- b. A student gives a check for \$10 to buy a used book. Later the student finds out that the check was cashed for \$100.
- c. A student sends hundreds of emails per day to the school using a phony return email address.

a. This is **snooping** (an attack to confidentiality). Although the contents of the test are not confidential on the day of the test, they are confidential before the test day.

b. This is **modification** (an attack to integrity). The value of the check is changed (from \$10 to \$100).

c. This is **denial of service** (an attack to availability). Sending so many emails may crash the server and the service may be interrupted.

Q2. Use the additive cipher with $k=10$ to encrypt the plaintext 'book'. Then decrypt the message to get the original plaintext.

- a. The following shows the character-by-character encryption of the plaintext. The ciphertext is "LYYU".

Plaintext	Encryption	Ciphertext
b $\rightarrow$ 01	$(01 + 10) \bmod 26$	11 $\rightarrow$ L
o $\rightarrow$ 14	$(14 + 10) \bmod 26$	24 $\rightarrow$ Y
o $\rightarrow$ 14	$(14 + 10) \bmod 26$	24 $\rightarrow$ Y
k $\rightarrow$ 10	$(10 + 10) \bmod 26$	20 $\rightarrow$ U

The following shows the character-by-character decryption of the ciphertext. The plaintext is "book".

Ciphertext	Decryption	Plaintext
L $\rightarrow$ 11	$(11 - 10) \bmod 26$	01 $\rightarrow$ b
Y $\rightarrow$ 24	$(24 - 10) \bmod 26$	14 $\rightarrow$ o
Y $\rightarrow$ 24	$(24 - 10) \bmod 26$	14 $\rightarrow$ o
U $\rightarrow$ 20	$(20 - 10) \bmod 26$	10 $\rightarrow$ k

Q3. Encrypt the message 'this is an exercise' using additive cipher with key = 20. Ignore the space between words. Decrypt the message to get the original plaintext.

The ciphertext is **NBCMCMUHYRYLWCMY**

Q4. *Atbash* was a popular cipher among Biblical writers. In *Atbash*, 'A' is encrypted as 'Z', 'B' is encrypted as 'Y', and so on. Similarly, 'Z' is encrypted as 'A', 'Y' is encrypted as 'B', and so on. Suppose that the alphabet is divided into halves and the letters in the first half are encrypted as the letters in the second and *vice versa*. Find the type of cipher and key. Encipher the plaintext 'an exercise' using the *Atbash* cipher.

The relationship between the plaintext and ciphertext is $P + C = 25 \bmod 26$. In other words, it is monoalphabetic cipher. However, the key is not fixed, it is a stream of odd integers as shown below (P is plaintext, K is key, and C is Cipher text):

P	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
K	25	23	21	19	17	15	13	11	09	07	05	03	01	25	23	21	19	17	15	13	11	09	07	05	03	01
C	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

Using the above relationship, we can encipher the plaintext as shown below:

Plaintext: an exercise

Ciphertext: ZMVCVIXRHV

Q5. Alice can use only the additive cipher on her computer to send a message to a friend. She thinks that the message is more secure if she encrypts the message two times, each time with a different key. Is she right? Defend your answer.

Double encryption here does not help. Encryption with k_1 followed by encryption with k_2 is the same as encryption with $k = (k_1 + k_2) \bmod 26$. Alice has only changed the key. The following shows the proof. P is the plaintext and C is the ciphertext.

$$C = [(P + k_1) \bmod 26 + k_2] \bmod 26$$

$$C = (P \bmod 26 + k_1 \bmod 26 + k_2) \bmod 26$$

$$C = (P \bmod 26) \bmod 26 + (k_1 \bmod 26) \bmod 26 + k_2 \bmod 26$$

$$C = P \bmod 26 + k_1 \bmod 26 + k_2 \bmod 26 = (P + k_1 + k_2) \bmod 26$$

$$C = (P + k) \bmod 26 \text{ where } k = (k_1 + k_2) \bmod 26$$

Q6. One of the attacks an intruder can apply to a simple cipher like an additive cipher is called the *ciphertext* attack. In this type of attack, the intruder intercepts the cipher and tries to find the key and eventually the plaintext. One of the methods used in a ciphertext attack is called the *brute-force* approach, in which the intruder tries several keys and decrypts the message until the message makes sense. Assume the intruder has intercepted the ciphertext 'UVACLYZLJBYL'. Try to decrypt the message by using keys from 1 until a plaintext appears that makes sense.

We apply the key = 1, 2, 3, 4, 5, 6, and 7 to find the plaintext.

Ciphertext	Key	Plaintext
UVACLYZLJBL	1	tuzbkxeykiaxy
UVACLYZLJBL	2	styajwdxjhzwj
UVACLYZLJBL	3	rsxzivcwigyvi
UVACLYZLJBL	4	qrwyhubvhfxuh
UVACLYZLJBL	5	pqvvgtaugewtg
UVACLYZLJBL	6	opuwfsztfdvsf
UVACLYZLJBL	7	notverysecure

Q7.

- What is the maximum number of keys that an eavesdropper needs to try in order to unscramble a ciphertext which has been encrypted using the Caesar Cipher?
- What would your answer be if a randomly-ordered cipher alphabet was employed?
- How could we try and crack a cipher based on a randomly-ordered cipher alphabet?

- All possible keys of a Caesar cipher can be obtained by shifting the plain alphabet 26 times.
- If a substitution cipher is used, an eavesdropper has to try up to: (explain the logic behind factorials)

$26 \times 25 \times 24 \times \dots \times 2 \times 1 = 26! = 403,291,461,126,605,635,584,000,000$ combinations!

- (iii) Frequency analysis can be used to identify the most popular letters in the ciphertext and map them to equally popular letters of the plain alphabet. Grammar and syntax rules of the adopted language can then be used to unscramble the remaining ciphertext.

Q8. A small private club has only 100 members:

How many secret keys are needed if all members of the club need to send secret messages to each other?

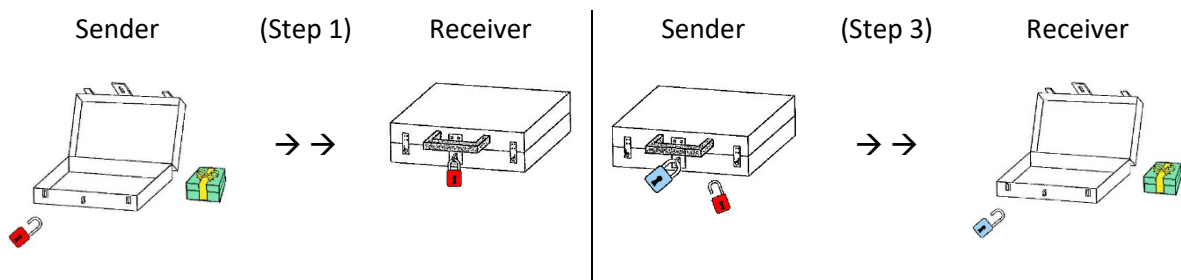
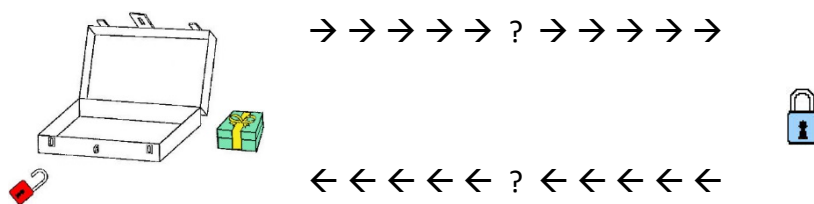
How many secret keys are needed if everyone trusts the president of the club? If a member needs to send a message to another member, she first sends it to the president; the president then sends the message to the other member.

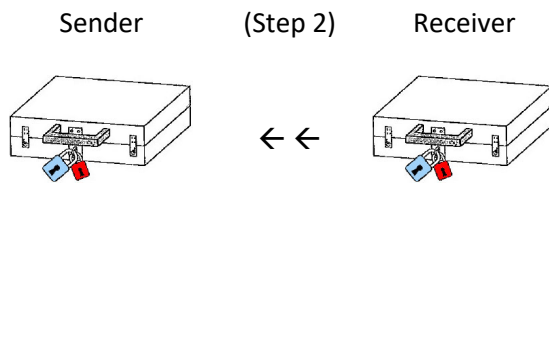
1. The number of keys = $n \times (n - 1) / 2 = (100 \times 99) / 2 = 4950$.
2. Only 100 keys are needed: There should be one secret key between the president and each member.
3. Only 100 keys are needed: There should be one secret key between the president and each member to create the session secret key. After the session key is established, the two members can use the one-time session key.

Q9. Symmetric-key cryptography is analogous to the exchange of a briefcase that contains a gift. The sender uses a padlock to lock the briefcase and hires a courier to deliver it to the intended receiver. A copy of the key is also sent to the receiver via a secure channel. Consider the case of asymmetric-key cryptography, where the sender owns a (red) padlock and a key, whilst the receiver owns a different (blue) padlock and a key. How should the sender have the briefcase delivered to the receiver without the requirement of key exchange?

Sender

Receiver





Alternative Notation Use $R(m)$ to denote that packet m is in a briefcase () and locked with a red padlock R . The sequence of events will be:

$m, R(m), R(m)B, R^{-1}R(m)B, (m)BB^{-1}, m$

Q10. Use the Playfair cipher to encipher the message "The key is hidden under the door pad". The secret key can be made by filling the first and part of the second row with the word "GUIDANCE" and filling the rest of the matrix with the rest of the alphabet.

Hint:

G	U	I/J	D	A
N	C	E	B	F
H	K	L	M	O
P	Q	R	S	T
V	W	X	Y	Z

Please note that

1. If the two letters in the pair are located in the same row of the secret key, the corresponding encrypted character for each letter is the next letter to the right in the same row (in case of last char- wrapping to the beginning of the row)
2. If the two letters in the pair are located in the same column of the secret key, the corresponding encrypted character for each letter is the letter beneath it, in the same column (with wrapping if the last one).
3. If the two letters in the pair are not in the same row or column of the secret key, the corresponding encrypted character for each letter is a letter that is in its own row but in the same column as the other letter.

The key is hidden under the door pad

PO CL BX DR LG IY IB CG BG LX PO BI LZ LT TG IY